International Institute of Medical Science and Technology council

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A

Mini Project Report

On

"AquaLife Coastal Watch"

(SDG 14 – Life Below Water)

Submitted By

Cohort No: 05

ABSTRACT

Marine ecosystems are facing increasing threats due to pollution, plastic waste, and illegal dumping in coastal regions. Traditional monitoring methods rely heavily on manual surveys and government reports, which often fail to capture real-time data from large coastal areas. This lack of real-time monitoring makes it difficult for environmental authorities to identify pollution hotspots and protect marine biodiversity.

The **AquaLife Coastal Watch system** is a citizen science web-based platform that allows users to report marine pollution and wildlife sightings. Citizens can upload geo-tagged images, provide location data, and describe environmental issues. The system collects this data and displays it on a map-based analytics dashboard for environmental agencies.

The platform improves marine conservation efforts by enabling community participation, providing real-time data, and supporting faster intervention strategies.

Technologies Used

- React.js (Frontend)
- Node.js & Express.js (Backend)
- MongoDB (Database)
- Google Maps API (Location tracking)
- JWT (Authentication)
- Cloud Storage (File upload)

Outcome

The system helps identify coastal pollution hotspots and promotes collaborative marine conservation.

CHAPTER 1: INTRODUCTION

1.1 Background

Marine ecosystems play a critical role in maintaining ecological balance and supporting biodiversity. However, coastal environments are increasingly threatened by pollution, plastic waste, oil spills, and habitat destruction.

Monitoring marine pollution traditionally depends on government inspections and environmental surveys. These methods are often slow, expensive, and unable to provide real-time data from large coastal areas.

Citizen science platforms allow communities to contribute environmental data through digital reporting tools. By enabling citizens to report marine pollution and wildlife sightings, authorities can gain real-time insights into environmental conditions.

The **AquaLife Coastal Watch** platform leverages digital technology and community participation to monitor coastal environments effectively.

1.2 Problem Statement

Marine conservation organizations lack real-time crowd-sourced data on coastal pollution hotspots and endangered marine species sightings. Without accurate and timely data, it becomes difficult to deploy targeted interventions and respond quickly to environmental threats.

1.3 Objectives

The main objectives of AquaLife Coastal Watch are:

- To enable citizens to report marine pollution incidents.
- To collect geo-tagged environmental data.
- To display pollution hotspots using map-based visualization.
- To assist environmental agencies in decision making.
- To promote public awareness about marine conservation.

1.4 Scope of the Project

The scope of the project includes:

- Web-based citizen reporting system
- Geo-tagged image uploads
- Marine pollution hotspot mapping
- Wildlife sighting reports
- Administrative dashboard for data validation
- Environmental analytics visualization

The system can be used by environmental agencies, NGOs, and coastal monitoring organizations.

CHAPTER 2: LITERATURE SURVEY

Marine environmental monitoring has traditionally relied on scientific surveys, laboratory testing, and satellite observations. These methods help detect pollution levels and environmental changes in oceans and coastal areas. However, such systems often require specialized equipment, trained personnel, and significant financial resources, which limits their large-scale implementation.

Traditional marine monitoring systems often face the following challenges:

- Limited real-time data
- High operational cost
- Lack of public engagement
- Difficulty monitoring large coastal areas

Some modern monitoring systems use IoT sensors to measure water quality parameters such as temperature, pH, and chemical contamination. While these systems provide real-time environmental data, they are expensive to install and maintain across large coastal regions.

In recent years, citizen science platforms have gained popularity in environmental monitoring. These platforms allow citizens to report environmental issues such as pollution, wildlife sightings, and waste accumulation using mobile applications or web-based systems. Citizen participation helps collect large amounts of data from different locations and improves environmental awareness.

However, many existing systems lack proper data verification, visualization tools, and centralized dashboards for authorities. The AquaLife Coastal Watch system addresses these limitations by integrating citizen reporting, geo-tagged data collection, map-based visualization, and administrative verification. This approach enhances real-time monitoring and supports better decision-making for marine conservation.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Existing System

The existing marine monitoring system involves:

- Government environmental surveys
- Manual pollution reporting
- Laboratory testing
- Limited digital monitoring systems

Problems in the existing system include:

- Lack of real-time reporting
- Slow response time
- Limited geographic coverage
- Lack of citizen participation

3.2 Proposed System

The proposed AquaLife Coastal Watch system provides a digital platform where citizens can report marine environmental issues using a web application.

The system:

- Collects geo-tagged images of pollution or wildlife sightings
- Stores reports in a centralized database
- Displays pollution hotspots using map visualization
- Allows moderators to verify reports
- Generates alerts for severe environmental threats

3.3 Feasibility Study

Technical Feasibility

The project uses widely available technologies such as React, Node.js, and MongoDB. These tools are stable, scalable, and suitable for web application development.

Economic Feasibility

The system uses open-source technologies, reducing development costs. Cloud hosting costs are minimal and scalable based on usage.

Operational Feasibility

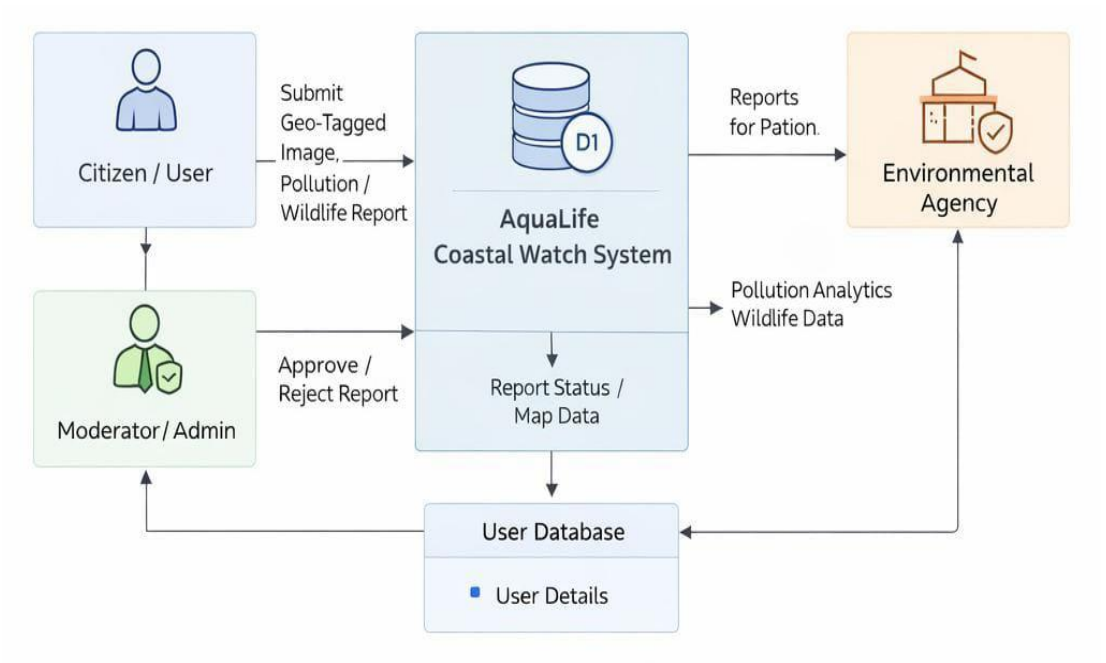
The system is user-friendly and requires minimal training. Community members can easily submit reports through a simple interface.

CHAPTER 4: SYSTEM DESIGN

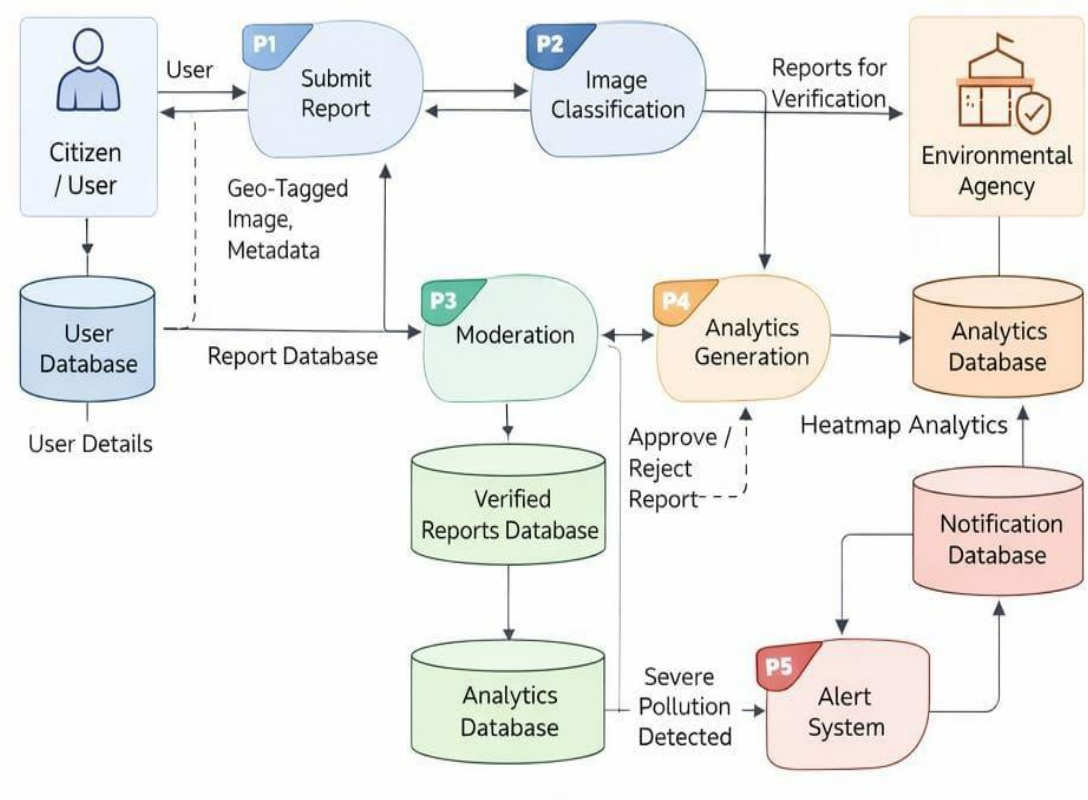
4.1 Overall System Architecture

4.2 Data Flow Diagram

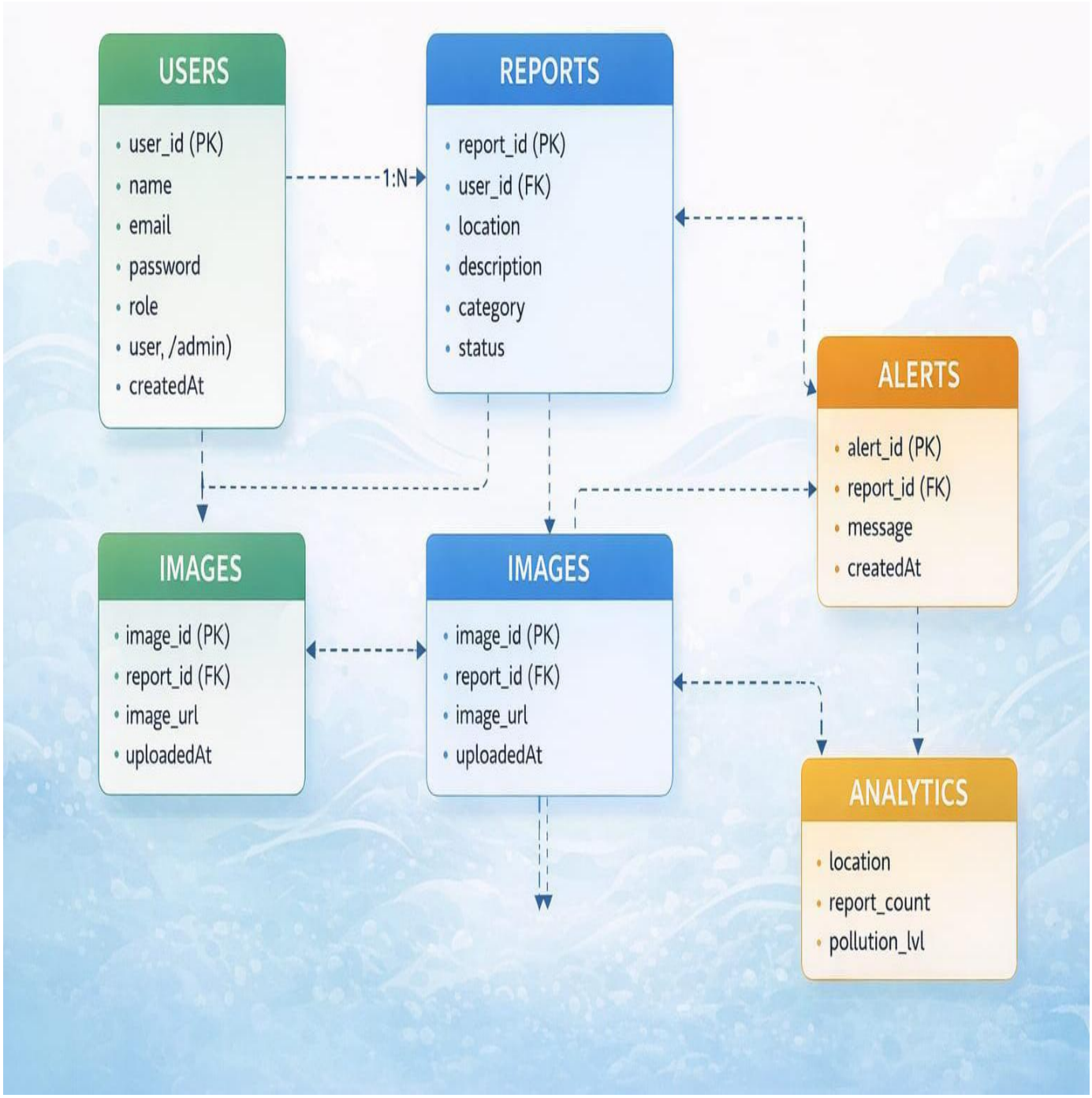
DFD Level 0



DFD Level 1



4.3 Database Design



CHAPTER 5: IMPLEMENTATION

5.1 Frontend Implementation

Developed using React.js with components such as:

- Login / Register Component
- Home Dashboard Component
- Coastal Map Component
- Pollution Report Form
- Image Upload Component
- Wildlife Sighting Component
- Admin Dashboard Panel
- Report Verification Panel

5.2 Backend Implementation

Built using Express APIs:

- /register-user
- /login-user
- /submit-report
- /get-reports
- /verify-report
- /update-report-status
- /heatmap-data

JWT is used for secure authentication.

5.3 Database Implementation

MongoDB schemas are defined for:

- Users
- PollutionReports
- WildlifeSightings
- Notifications

Mongoose is used to connect Node.js with MongoDB.

5.4 API Endpoints

Examples:

POST /api/register

POST /api/login

POST /api/report

GET /api/reports

GET /api/heatmap

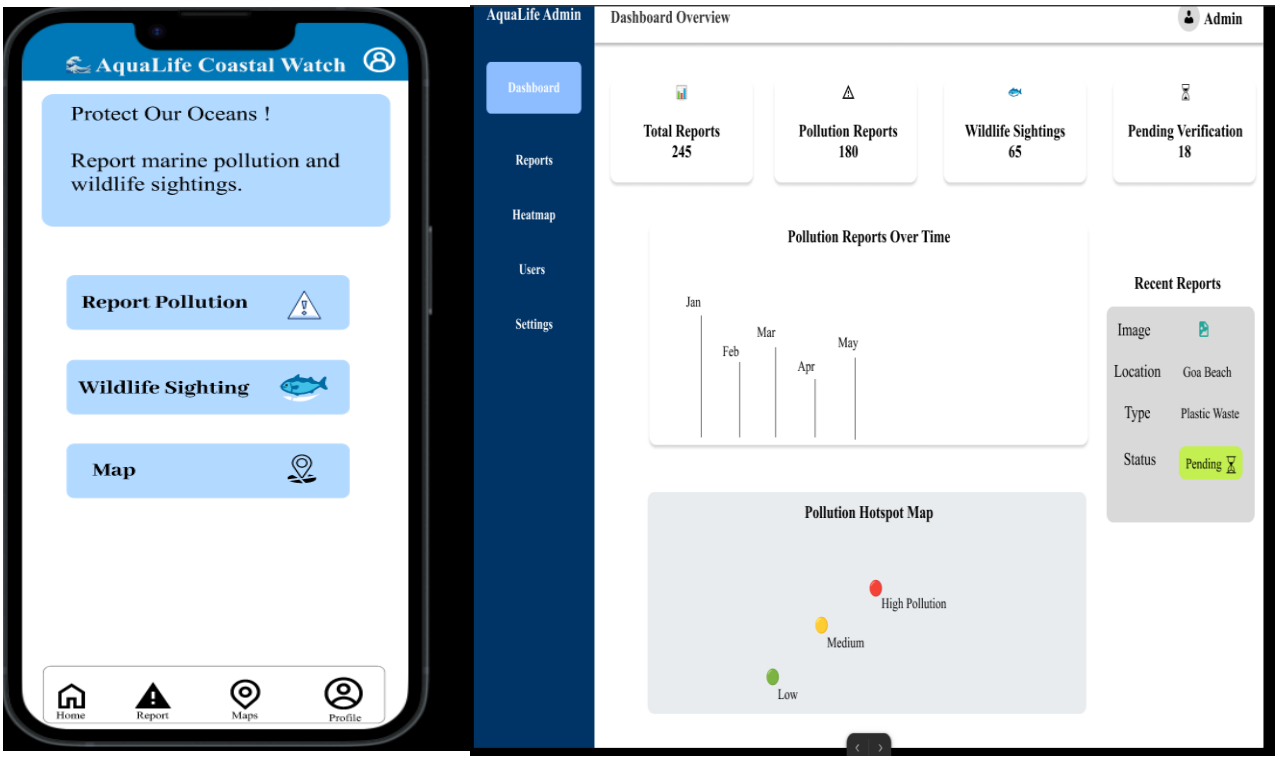
PUT /api/report/:id

PUT /api/report/status

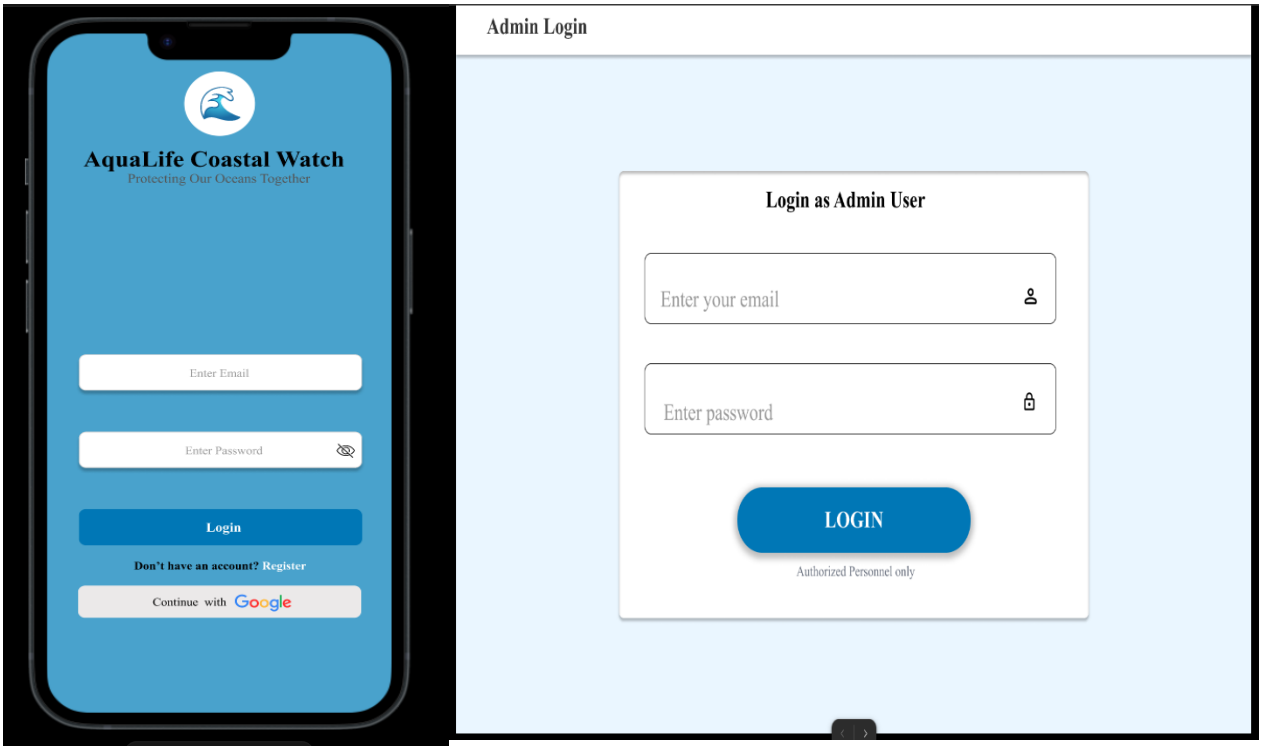
CHAPTER 6: RESULTS AND OUTPUT

The system successfully provides:

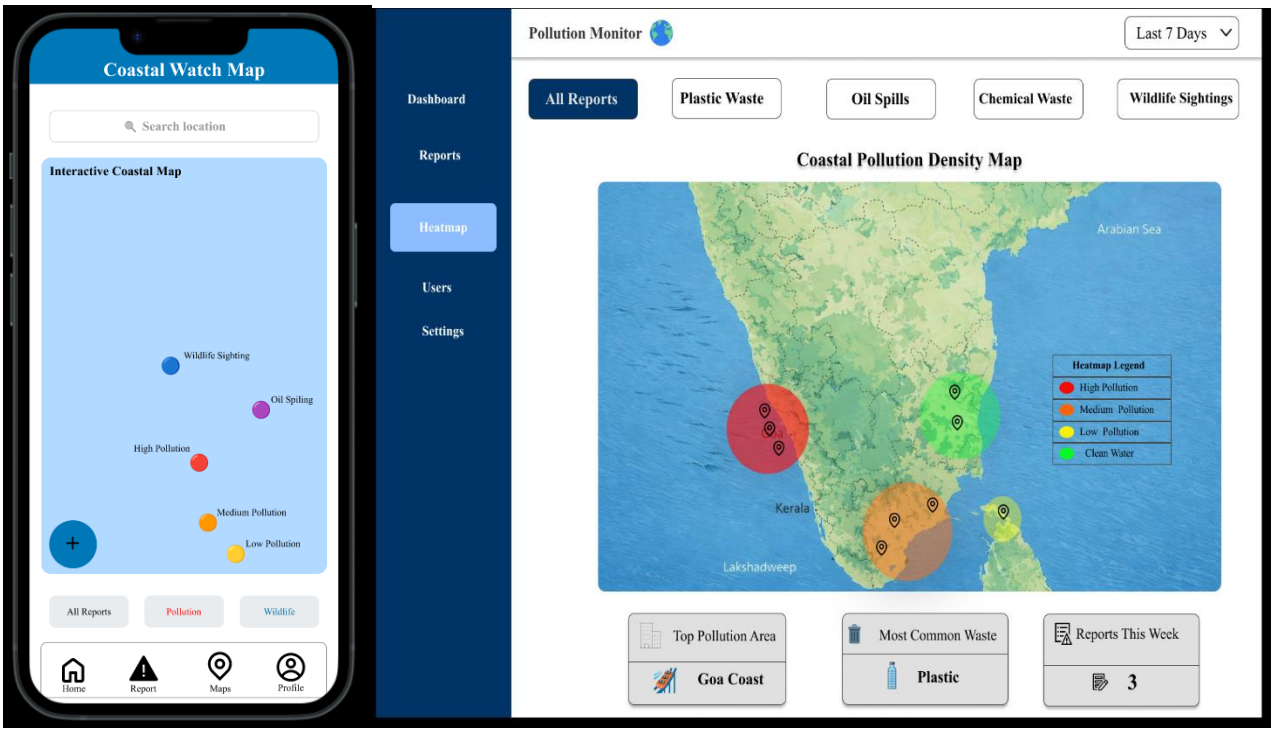
6.1 User Dashboard and Admin Dashboard



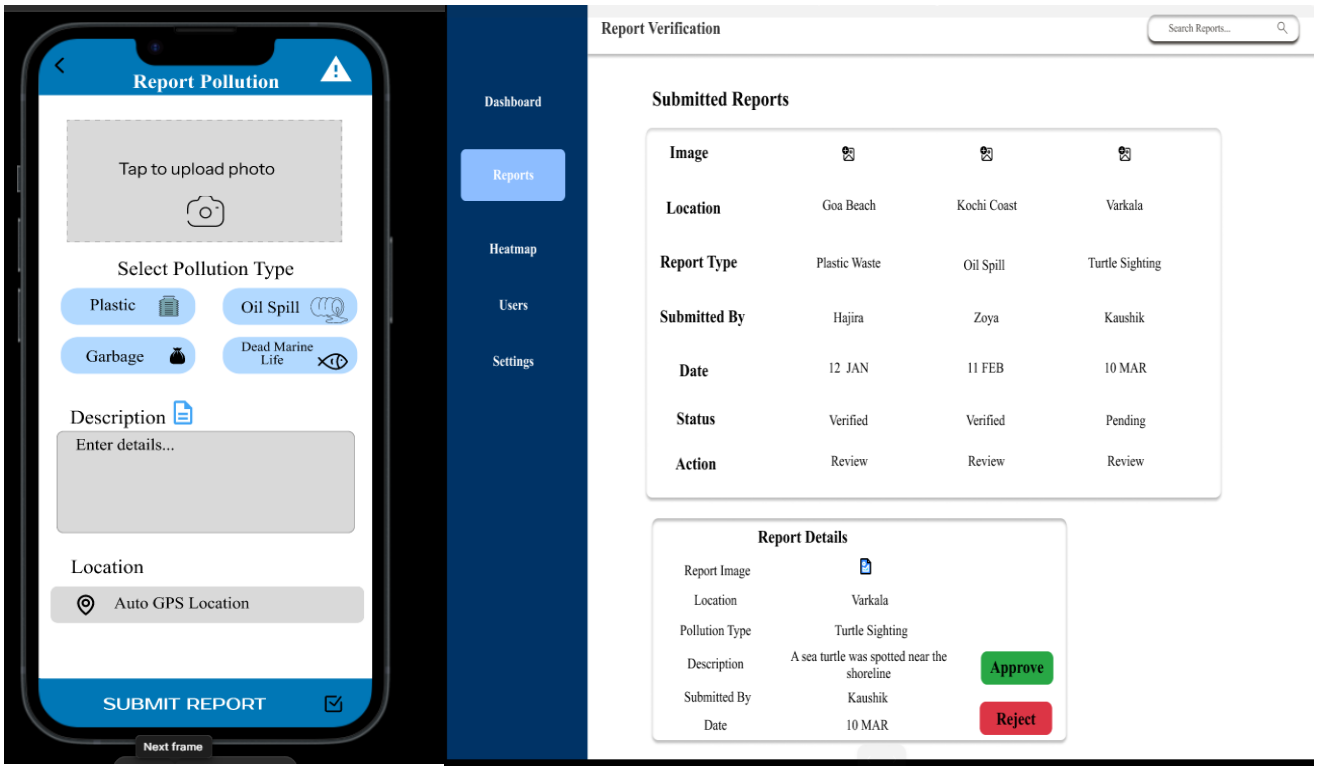
6.2 Logging Page for Both User and Admin Panel



6.3 Heat Map



6.4 Report Submission and Verification page



CHAPTER 7: SDG Impact Analysis

SDG 14: Life Below Water

Goal Description:

SDG 14 focuses on conserving and sustainably using oceans, seas, and marine resources for sustainable development. Marine ecosystems are critical for biodiversity, climate regulation, and economic activities such as fishing and tourism.

Contribution of AquaLife Coastal Watch

The AquaLife Coastal Watch system directly supports SDG 14 by enabling monitoring and reporting of coastal environmental issues. Citizens can report marine pollution events, plastic waste accumulation, oil spills, and wildlife sightings through the platform.

The system collects geo-tagged environmental data and displays pollution hotspots on an interactive map. This information helps environmental agencies identify areas that require immediate attention.

Key contributions include:

- Monitoring marine pollution through community reporting
- Identifying coastal pollution hotspots
- Supporting protection of endangered marine species
- Encouraging public participation in marine conservation

Impact:

By enabling real-time environmental reporting, the platform improves the ability of authorities to respond quickly to pollution incidents. This helps protect marine biodiversity and promotes sustainable ocean management.

CHAPTER 8: CONCLUSION

The AquaLife Coastal Watch system provides an effective digital solution for monitoring coastal environmental issues. By enabling community participation and integrating geospatial analytics, the system improves marine conservation efforts.
The project demonstrates how citizen science platforms can contribute to environmental protection and support sustainable development goals.

CHAPTER 9: FUTURE ENHANCEMENTS

Future improvements may include:

- AI-based marine pollution detection
- Drone-based coastal monitoring
- IoT sensor integration
- Mobile application version
- Advanced environmental analytics